More women owners, or more women working at home on contract? Women's enterprise ownership in India, 2015–16 to 2023–24

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Abstract. The share of India's unincorporated non-agricultural enterprises owned by women rose sharply in the last decade. Using unit-level data from the NSS 73rd round (2015–16) and the Annual Survey of Unincorporated Sector Enterprises 2023–24, we ask where the new women-owned enterprises are and what they earn. We decompose the change in the number of female-proprietor enterprises into cells defined by activity, home location, dependence on a contractor and rural or urban area, split the change in the female share into within- and between-cell parts, and compare value added per worker with women's earnings outside the enterprise, estimated from the Periodic Labour Force Survey 2023–24. The female-proprietor share in a harmonised coverage rose from 19.5 to 26.3 per cent, driven by changes within cells. Home-based enterprises selling on their own account account for 65.3 per cent of the net rise in women owners, while the share of female-proprietor enterprises working mainly for a contractor fell from 31.2 to 17.0 per cent. Returns remain low: the median female-proprietor enterprise generates Rs 3,558 of value added per worker per month, and 80.1 per cent generate less than 25 days of women's casual wage, partly because they operate fewer hours. The net growth is in enterprises more independent than before, but most are small home-based units with low returns; ownership counts should be read alongside location and returns.

Keywords: women's entrepreneurship; home-based work; putting-out; informal enterprises; India

JEL codes: J16, L26, O17, J22

Plain English Summary. More of India's small non-farm businesses are now owned by women. We use two large government surveys, from 2015–16 and 2023–24, to see what kind of businesses these are. Most of the net growth is in very small businesses run from home. Fewer of them now work for a contractor who supplies the work and buys the output, so in that sense they have become more independent. But most of them still produce less value per worker than a woman would earn in a month of daily wage labour, partly because they work fewer hours. The number of women business owners is a weak guide to women's economic progress. Policy that counts women owners should also track where they work, who they work for, and what they earn.

1 Introduction

India's official enterprise surveys show a fast rise in women's ownership of unincorporated non-agricultural enterprises. In the NSS 73rd round (2015–16), 19.5 per cent of enterprises in our harmonised coverage were owned by a woman proprietor; in the Annual Survey of Unincorporated Sector Enterprises (ASUSE) 2023–24 the figure was 26.3 per cent (National Sample Survey Office, 2017; National Statistics Office, 2024b). The published factsheet for 2023–24, whose coverage differs slightly from ours (Section 2.3), reports a rise from 22.9 per cent in 2022–23 to 26.2 per cent in a single year (National Statistics Office, 2024b). Because these surveys are the main source on India's informal enterprises, the rise is easily read as evidence of growing women's entrepreneurship.

This paper asks a simple question: where are the new women-owned enterprises, and what do they earn? Ownership in these surveys is a statement about who runs the enterprise; it says nothing about whether the enterprise sells on its own account, is run from a separate workplace, or earns more than the owner's alternatives. A woman who stitches garments at home for a contractor who supplies the cloth and buys every piece is recorded as the

proprietor of an own-account manufacturing enterprise. So is a woman who runs a shop. The two are different economic situations, and the literature on home-based and subcontracted work has long argued that the first is closer to dependent wage work than to entrepreneurship (Chen, Sebstad, and O'Connell, 1999; Kesar, 2024).

We make three contributions. First, we bring the two surveys onto a common coverage and decompose the change in the number of female-proprietor enterprises into cells defined by broad activity, location of the enterprise (in the owner's home or outside), dependence on a contractor (work done solely or mainly on contract) and rural or urban area. A shift-share split separates growth of women's ownership within cells from reallocation of enterprises across cells. Second, we profile the cells that account for most of the net rise. Because the surveys are repeated cross-sections, these are net changes in the stock of enterprises, not the characteristics of new entrants. Third, we benchmark the returns of these enterprises against the outside option of the women who run them, the earnings of women casual workers and self-employed women in the Periodic Labour Force Survey (PLFS) 2023–24 (National Statistics Office, 2024a). This benchmark turns a statement about ownership into a statement about welfare: an enterprise that generates less value added per worker than the casual wage is a poor route out of low-paid work, whatever its legal form.

The closest studies are Bose (2023), who documents a large productivity penalty for women-owned home-based manufacturing firms in the 2015–16 data, Kesar (2024), who shows that women-owned informal enterprises are far more likely to work under subcontract, and Mohanan and Kar (2025), who link the recent growth of unorganised manufacturing to women's job work. Shuvam and Mohanty (2023) classify proprietorship by necessity and opportunity motives across earlier rounds. To our knowledge, none of these studies follows the change from 2015–16 to the post-pandemic ASUSE rounds across manufacturing, trade and services, decomposes it by home location and contract dependence together, or compares enterprise returns with women's outside option. The paper also speaks to work on gender gaps in microenterprise returns (de Mel, McKenzie, and Woodruff, 2009; Fafchamps et al., 2014), on credit and ownership in India (Chaudhuri, Sasidharan, and Raj, 2020), on the local determinants of female entrepreneurship (Ghani, Kerr, and O'Connell, 2013) and on the stagnation

and measurement of women's work in India (Deshpande and Kabeer, 2021; Klasen and Pieters, 2015).

The answer is mixed. The rise in women's ownership does not appear to be a rise in putting-out, at least as the surveys record contract work: of the 6.7 point increase in the female share, 9.2 points occurred within cells, while the shrinking weight of home-based contract manufacturing, where women's ownership was already high, pulled the share down by 2.5 points. The number of female-proprietor enterprises working mainly for a contractor fell from 38.6 lakh to 32.5 lakh. But the net growth is concentrated in home-based enterprises selling on their own account, which account for 65.3 per cent of the rise, and these generate far less value added per worker than men's enterprises or than casual wage work.

2 Data

2.1 Enterprise surveys

The NSS 73rd round (July 2015 to June 2016, Schedule 2.34) and ASUSE 2023–24 (October 2023 to September 2024) are nationally representative surveys of unincorporated non-agricultural enterprises, excluding construction, run by the National Statistics Office with a two-stage stratified design (National Sample Survey Office, 2017; National Statistics Office, 2024b). ASUSE uses the same concepts as the earlier rounds and a closely similar schedule, and the items we use are asked in both (we map the codes item by item in the replication code): type of ownership (with separate codes for male, female and transgender proprietors), location of the enterprise (within household premises or outside), whether the enterprise is an own-account enterprise (OAE, no regular hired worker) or an establishment, whether it works on contract and, if so, whether solely or mainly for a contractor or mainly for its own customers, whether the contractor supplies raw material or equipment, months of operation, usual hours a day, and the number of working owners, hired workers and unpaid helpers by sex. The released NSS 73rd round file holds 290,113 enterprise records and the ASUSE 2023–24 file 523,775 establishment records (the latter including residential renting); after

harmonising coverage we keep 289,843 and 495,451 records.

One difference in the questionnaires matters for the contract variable. The NSS 73rd round asked every enterprise whether it worked on contract. In ASUSE 2023–24 the question for manufacturing is asked only of establishments that report providing manufacturing services, that is, transforming inputs they do not own, and custom work for households is not treated as contract work; a separate item covers contract work in other services. Among women's home-based manufacturing establishments in ASUSE 2023–24, 94.4 per cent report providing manufacturing services, so the routing itself can affect only a small share of them, but the narrower guidance may record fewer contract arrangements than the 2015–16 wording. We return to this in the Discussion.

We use the final multipliers supplied with each data set, the average of the two semi-round multipliers for the 2015–16 round as prescribed in its estimation procedure, and the stratum and first-stage unit identifiers for the design-based standard errors. To compare like with like we restrict both rounds to manufacturing (NIC 2008 divisions 10–33), trade (45–47) and other services (49–96 except 84), and drop residential renting (NIC 68108), which ASUSE tabulates separately. "Female-proprietor enterprise" means proprietary ownership by a woman; partnerships, self-help groups, cooperatives and other forms are kept as a third category.

2.2 Returns and the outside option

Gross value added (GVA) is the survey's own item, computed from receipts and operating expenses for the reference period and annualised following the published procedure (by twelve for perennial enterprises and by the reported months of operation for seasonal and casual ones). We work with monthly GVA per worker, where workers include working owners and unpaid helpers, and restrict the returns analysis to market enterprises. GVA per worker is an upper bound on the owner's earning per worker, because it is before rent, interest and the cost of capital.

The outside option comes from the PLFS 2023–24 unit data. We take the average daily

earnings of women in casual wage work in current weekly status (Rs 291 in rural areas and Rs 359 in urban areas; Rs 298 overall, against Rs 450 for men), and the average gross earnings over the last 30 days of self-employed women (Rs 5,497, against Rs 16,150 for men). A month of casual work at 25 days is therefore worth Rs 7,438; we report 22 and 30 days as sensitivity bounds.

2.3 Validation

Before any analysis, we reproduce from the unit data each estimate that has a published counterpart. The PLFS benchmark reproduces the published quarterly all-India average wage of women casual workers in current weekly status within Rs 1.8 a day in rural areas and within Rs 7.5 in urban areas in every quarter (Rs 2.1 for India as a whole), the male all-India wage within Rs 0.7, and the published 30-day earnings of self-employed women within 0.6 per cent (Table 5 in the Appendix). For the enterprise surveys we reproduce the number of enterprises, the female-proprietor share, GVA per worker and GVA per enterprise for each round (Table 6 in the Appendix): 13 published statistics in all, including the number of workers and total GVA. The largest gap is 0.45 per cent, and every enterprise-survey gap is within the rounding of the published figure. Two definitional points matter. First, the published female-owned share for 2023–24 is reproduced only when female proprietary establishments are divided by all establishments (26.17 per cent); as a share of proprietary establishments alone the figure is 27.64 per cent (19.54 and 20.36 per cent in 2015–16). We use this denominator (all enterprises, not only proprietary ones) throughout. Restricted to our harmonised industry coverage, the 2023–24 share is the 26.25 per cent used in the analysis, against 19.55 per cent in 2015–16. Second, GVA in both rounds is reproduced only for market enterprises, that is excluding non-profit institutions and head offices (NIC 70100), with the reference-period value annualised as described above.

3 Method

Let c index cells formed by activity (manufacturing, trade, other services), location (home, outside), contract dependence (solely or mainly on contract, labelled "contract"; otherwise, labelled "own-market") and area (rural, urban), N_{ct} the weighted number of enterprises in cell c in year t, F_{ct} the number with a female proprietor, $e_{ct} = N_{ct}/\sum_c N_{ct}$ the cell share and $s_{ct} = F_{ct}/N_{ct}$ the female share.

The *count decomposition* reports $\Delta F_c/\sum_c \Delta F_c$, the share of the rise in women owners that each cell accounts for. The *shift-share decomposition* of the change in the overall female share $S_t = \sum_c e_{ct} s_{ct}$ is

$$\Delta S = \sum_c \bar{e}_c \, \Delta s_c + \sum_c \bar{s}_c \, \Delta e_c,$$

with bars denoting the average of the two years; the first term is the within-cell change and the second the change due to growth of cells where women's ownership is high. If the rise were driven by the growth of home-based contract work, the between term and the contract cells would carry it; a large within term together with a rising contribution of own-market cells points instead to more women owners within each type of enterprise. The decomposition is descriptive and cannot by itself distinguish new entry by women from changes in whose name an existing household enterprise is recorded in.

For returns, we report the weighted median and quartiles of monthly GVA per worker by ownership type, home location and contract dependence, the share of enterprises whose GVA per worker falls below 22, 25 and 30 days of the woman casual wage, and GVA per worker-hour. We then estimate, with survey weights and design-based standard errors, regressions of log GVA per worker on a female-proprietor indicator, urban location, NIC two-digit and state fixed effects (model 1), adding home location, contract dependence, own-account status, log hours a day and log months a year (model 2), and interactions of the female indicator with home and contract (model 3). The change in the female coefficient from model 1 to model 2 shows how much of the raw gap is associated with where and how women's enterprises operate. Hours, months and own-account status may themselves reflect the constraints women face, and annual GVA is built from months of operation, so model 2 is descriptive;

none of the models identifies the effect of ownership on returns. Enterprises with zero or negative GVA (296 sample enterprises) are kept in the medians and shares but dropped from the log regressions. In the text we convert coefficients to per cent differences, $100(e^b - 1)$; Table 4 reports log points.

4 Results

4.1 Where the rise came from

Table 1: Cells accounting for the change in the number of female-proprietor enterprises, 2015–16 to 2023–24 (ten largest)

Cell	2015–16 (lakh)	2023–24 (lakh)	Change (lakh)	Share of change (%)
Manufacturing | home | own-market | rural	25.8	50.2	24.4	35.8
Other services | outside | own-market | urban	3.9	18.8	14.9	21.9
Manufacturing | home | own-market | urban	20.7	28.5	7.7	11.4
Other services | home | own-market | rural	2.9	7.9	5.0	7.3
Trade | outside | own-market | rural	3.8	8.0	4.2	6.1
Other services | outside | own-market | rural	1.4	5.3	4.0	5.8
Other services | home | own-market | urban	7.1	11.0	4.0	5.8
Trade | outside | own-market | urban	5.8	9.4	3.6	5.3
Trade | home | own-market | rural	5.8	8.9	3.1	4.6
Manufacturing | outside | own-market | rural	1.4	3.4	2.0	3.0

Table 1 lists the ten cells that account for most of the change. The number of female-proprietor enterprises in the harmonised coverage rose from 123.8 lakh to 191.9 lakh. The largest single cell is Manufacturing | home | own-market | rural, which accounts for 35.8 per cent of the rise. Taken together, home-based cells account for 56.9 per cent and contract cells for -8.9 per cent: home-based cells selling on their own account contribute 65.3 per cent, offset in part by the fall in home-based contract manufacturing. In the shift-share split, the female share rose by 6.7 percentage points (from 19.55 to 26.25 per cent), of which 9.2 points is within cells and -2.5 points is due to the changing mix of cells.

4.2 Where the net growth sits

Table 2: Profile of proprietary enterprises by sex of owner. Lakh: number of enterprises; OAE, Home, Contract (any work on contract), Mainly contract (solely or mainly for a contractor) and Mfg (manufacturing) are percentages of enterprises; Workers: average per enterprise.

Year	Owner	Lakh	OAE	Home	Contract	Mainly contract	Mfg	Hours/day	Workers
2023-24	Female	191.9	96.7	74.8	17.4	17.0	61.3	5.8	1.24
2023-24	Male	500.1	83.0	23.4	2.6	2.2	16.5	8.4	1.77
2015-16	Female	123.8	96.1	84.7	32.3	31.2	71.5	6.1	1.31
2015-16	Male	484.0	81.4	33.3	4.7	3.9	21.6	8.5	1.80

Table 2 compares female- and male-proprietor enterprises in the two rounds. In 2023–24, 74.8 per cent of female-proprietor enterprises operated from the owner's home, against 23.4 per cent of male-proprietor enterprises, and 17.0 per cent worked solely or mainly on contract, against 2.2 per cent. The share of female-proprietor enterprises working mainly on contract fell from 31.2 per cent in 2015–16, and the share at home from 84.7 per cent. Women's enterprises also became more connected: 14.1 per cent used the internet in 2023–24, against 1.2 per cent in 2015–16. Among female own-account enterprises at home, the share mainly on contract changed from 36.4 to 22.5 per cent, and the share in manufacturing from 80.8 to 77.4 per cent.

4.3 Returns against the outside option

Table 3: Monthly GVA per worker (Rs), proprietary market enterprises, 2023–24. N: sample enterprises (* fewer than 100: imprecise). OAE: own-account enterprise; Contract: solely or mainly on contract. <22d, <25d, <30d: percentage of enterprises below 22, 25 and 30 days of the average daily wage of women casual workers (PLFS 2023–24). Per hour: GVA per worker-hour, assuming 25 working days a month.

Owner	OAE	Home	Contract	N	Median	P25	P75	<22d	<25d	<30d	Per hour
Female	All	All	All	108,805	3,558	1,983	6,355	75.8	80.1	85.2	38.1
Male	All	All	All	354,922	10,670	6,538	15,970	25.0	30.1	39.2	59.3
Female	Yes	No	No	27,340	6,221	3,755	10,020	52.6	59.5	69.6	41.1

Owner	OAE	Home	Contract	N	Median	P25	P75	<22d	<25d	<30d	Per hour
Female	Yes	No	Yes	123	3,305	1,447	4,810	87.7	92.6	95.2	27.9
Female	Yes	Yes	No	65,321	3,290	1,830	5,541	82.0	86.2	90.6	31.4
Female	Yes	Yes	Yes	8,466	2,230	1,318	3,180	97.8	98.7	99.4	17.8
Female	No	No	No	6,532	14,039	9,781	20,040	9.7	13.2	20.3	69.8
Female	No	No	Yes	84*	12,991	9,399	17,700	3.6	6.5	16.8	58.3
Female	No	Yes	No	903	9,085	6,268	13,382	27.0	35.6	47.9	55.1
Female	No	Yes	Yes	36*	8,266	6,808	8,266	22.1	31.5	82.4	27.6
Male	Yes	No	No	197,011	11,003	7,219	16,050	21.2	26.4	36.2	55.6
Male	Yes	No	Yes	651	9,740	6,440	14,150	25.2	36.0	46.4	46.1
Male	Yes	Yes	No	60,619	6,635	3,800	11,015	49.3	55.3	64.4	41.8
Male	Yes	Yes	Yes	2,172	4,928	2,625	8,285	65.1	69.7	78.1	25.2
Male	No	No	No	83,481	15,045	10,810	20,508	5.7	8.6	14.7	71.1
Male	No	No	Yes	2,569	13,625	10,468	17,665	3.1	6.0	13.2	59.0
Male	No	Yes	No	7,918	11,600	8,026	16,178	13.8	20.3	31.3	59.6
Male	No	Yes	Yes	501	10,692	7,912	13,950	15.1	21.4	36.7	46.7

The median female-proprietor enterprise generates Rs 3,558 of value added per worker per month, against Rs 10,670 for male proprietors and Rs 7,438 for 25 days of women's casual work at Rs 297.5 a day. 80.1 per cent of female-proprietor enterprises fall below that benchmark, against 30.1 per cent of male-proprietor enterprises (75.8 and 85.2 per cent of female-proprietor enterprises with 22- and 30-day months; Table 3). The gap is widest at home: among own-account enterprises that do not work mainly for a contractor, 86.2 per cent of women's home-based units fall below the benchmark, against 59.5 per cent of those outside the home, and women's home-based contract units have a median of only Rs 2,230. Part of the gap is time: female proprietors report 5.8 hours a day against 8.4 for men, and per worker-hour the home-based own-account gap is smaller (Rs 31.4 against Rs 41.8).

In the regressions (Table 4), GVA per worker in female-proprietor enterprises is 52.3 per cent lower than in male-proprietor enterprises in the same activity, state and area. Controlling for home location, contract dependence, own-account status, hours and months, the gap is 35.2 per cent. Working at home is associated with 23.7 per cent lower GVA per worker and contract dependence with 20.9 per cent lower. In model 3 both penalties are larger for

women (interaction coefficients, in log points, -0.117 and -0.127), and the female coefficient for an enterprise outside the home and not on contract is -0.361 log points.

Table 4: Log monthly GVA per worker, proprietary market enterprises, 2023–24 (N = 463,431). Survey-weighted; standard errors in parentheses account for strata and first-stage units. All models include NIC two-digit and state fixed effects.

Variable	(1)	(2)	(3)
Female proprietor	-0.741 (0.015)	-0.433 (0.013)	-0.361 (0.020)
Urban	0.331 (0.010)	0.236 (0.009)	0.234 (0.009)
At home		-0.270 (0.011)	-0.242 (0.011)
Mainly on contract		-0.235 (0.032)	-0.156 (0.049)
Own-account		-0.235 (0.008)	-0.237 (0.008)
Log hours a day		0.622 (0.012)	0.614 (0.012)
Log months a year		0.256 (0.027)	0.256 (0.027)
Female x home			-0.117 (0.022)
Female x contract			-0.127 (0.049)

5 Discussion

The results do not support the simplest pessimistic reading of the rise in women's ownership. Had the rise come from the spread of putting-out, with women recorded as proprietors of home-based units that stitch, roll or assemble for a contractor, the contract cells and the between-cell term would carry it. They do not. Contract cells account for -8.9 per cent of the change in the number of women owners, and the between-cell term is negative, because home-based contract manufacturing, the cell where women's ownership was highest in 2015–16, shrank as a share of all enterprises. The fall in contract dependence among women owners is large. It suggests that the subcontracting linkages documented for the earlier rounds (Kesar, 2024) have become less common among women's enterprises, at least in the forms the surveys record. That qualification matters: the contract item records work done solely or mainly for a contractor, and dependence on a single trader who buys the output or advances credit, common in putting-out, may be reported as sales to own customers. Together with the narrower ASUSE guidance noted in Section 2.1, part of the fall could

reflect how such arrangements are recorded rather than the arrangements themselves.

The results do not support the optimistic reading either. Most of the net rise in women-owned enterprises is in home-based units selling on their own account: the rural home-based manufacturing cell alone accounts for 35.8 per cent of the rise, and home-based own-market cells together for 65.3 per cent, against 43.6 per cent for own-market enterprises outside the home. Apart from home-based contract units, these are the units with the lowest returns in the data. Their median value added per worker is below what a woman casual worker earns in 25 days, and below the average 30-day earnings that self-employed women report in the PLFS (Rs 5,497). The gap with men persists after controlling for activity, state, location, contract status and time worked, in line with the home-based productivity penalty found for women-owned manufacturing firms in India (Bose, 2023); experiments in Sri Lanka and Ghana also find lower returns in women's microenterprises (de Mel, McKenzie, and Woodruff, 2009; Fafchamps et al., 2014).

Two readings fit these facts. One is that the enterprise surveys now capture more of the low-intensity, home-based production that women combine with domestic work, which the shorter hours of women's enterprises suggest; in that case the rise in ownership is partly a rise in measured, rather than actual, activity. The other is that more women have taken up independent self-employment because wage work close to home is scarce, so that the enterprise is a fallback rather than an opportunity, in the sense of Shuvam and Mohanty (2023). The data cannot separate these, nor can they tell whether some enterprises are recorded in a woman's name while being run jointly within the household. The counts make the question pressing: the number of male-proprietor enterprises changed little, from 484.0 lakh to 500.1 lakh, while the number of female-proprietor enterprises rose by more than half. Both readings imply that the count of women owners is a weak indicator of women's economic progress.

Three caveats apply. First, eight years separate the rounds, and the period includes the 2016 demonetisation, the introduction of the goods and services tax in 2017 and the pandemic; the comparison describes the net change and does not attribute it to any of them. The published one-year rise in the female-proprietor share between ASUSE 2022–23 and 2023–24

(22.9 to 26.2 per cent) also suggests that part of the change is recent, or that changes in the sampling frame or field practice between survey rounds contributed; the ASUSE 2021–22 and 2022–23 unit data could be used to check this. Second, GVA per worker is not the owner's income: it includes the return to capital and is shared with unpaid helpers. Third, the PLFS casual wage is the earning of women who found casual work, and may overstate what the average woman proprietor could earn outside her enterprise; it is an average over days actually worked and ignores days without work. Finally, we report design-based standard errors for the regressions but not for the decomposition terms, medians and shares below the benchmark, which should be read as point estimates.

6 Conclusion

Between 2015–16 and 2023–24 the share of India's unincorporated non-agricultural enterprises with a woman proprietor rose from 19.5 to 26.3 per cent in a common coverage. As far as the surveys record it, the rise was not driven by home-based contract work, which declined, but by home-based enterprises selling on their own account, most of which generate less value added per worker than casual wage work. For research, this means that ownership is a poor proxy for entrepreneurship in these surveys unless it is combined with location, contract status and returns, all of which the surveys already collect. Because GVA per worker is not the owner's income and the casual wage is an imperfect outside option, the share below the benchmark is indicative rather than exact. For policy, programmes that set targets in terms of the number of women-owned enterprises would do better to track the share of those enterprises that operate outside the home, sell on their own account and earn more than the local wage.

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References

Bose, S. (2023). The penalty of work from home: Gender gap in productivity of unorganised manufacturing firms in India. *Small Business Economics*, *60*(1), 351–369. https://doi.org/10.1007/s11187-022-00637-2

Chaudhuri, K., Sasidharan, S., and Raj, R. S. N. (2020). Gender, small firm ownership, and credit access: Some insights from India. *Small Business Economics*, *54*(4), 1165–1181. https://doi.org/10.1007/s11187-018-0124-3

Chen, M., Sebstad, J., and O'Connell, L. (1999). Counting the invisible workforce: The case of homebased workers. *World Development*, *27*(3), 603–610. https://doi.org/10.1016/S0305-750X(98)00154-5

de Mel, S., McKenzie, D., and Woodruff, C. (2009). Are women more credit constrained? Experimental evidence on gender and microenterprise returns. *American Economic Journal: Applied Economics*, *1*(3), 1–32. https://doi.org/10.1257/app.1.3.1

Deshpande, A., and Kabeer, N. (2021). *Norms that matter: Exploring the distribution of women's work between income generation, expenditure-saving, and unpaid domestic responsibilities in India* (WIDER Working Paper 2021/70). UNU-WIDER. https://doi.org/10.35188/unu-wider/2021/070-2

Fafchamps, M., McKenzie, D., Quinn, S., and Woodruff, C. (2014). Microenterprise growth and the flypaper effect: Evidence from a randomized experiment in Ghana. *Journal of Development Economics*, *106*, 211–226. https://doi.org/10.1016/j.jdeveco.2013.09.010

Ghani, E., Kerr, W. R., and O'Connell, S. D. (2013). Local industrial structures and female

entrepreneurship in India. *Journal of Economic Geography*, *13*(6), 929–964. https://doi.org/10.1093/jeg/lbt004

Kesar, S. (2024). Subcontracting linkages in India's informal economy. *Development and Change*, *55*(1), 38–75. https://doi.org/10.1111/dech.12817

Klasen, S., and Pieters, J. (2015). What explains the stagnation of female labor force participation in urban India? *The World Bank Economic Review*, *29*(3), 449–478. https://doi.org/10.1093/wber/lhv003

Mohanan, P. C., and Kar, A. (2025). Growth of unorganised manufacturing in India: Implications for women workers. *Review of Agrarian Studies*, *15*(1), 39–54. https://doi.org/10.25003/ras.15.01.0004

National Sample Survey Office. (2017). *Key indicators of unincorporated non-agricultural enterprises (excluding construction) in India, NSS 73rd round (July 2015–June 2016)*. Ministry of Statistics and Programme Implementation, Government of India.

National Statistics Office. (2024a). *Annual report, periodic labour force survey (PLFS), July 2023–June 2024*. Ministry of Statistics and Programme Implementation, Government of India.

National Statistics Office. (2024b). *Annual survey of unincorporated sector enterprises (ASUSE) 2023–24: factsheet*. Ministry of Statistics and Programme Implementation, Government of India.

Shuvam, S., and Mohanty, P. C. (2023). Necessity or opportunity driven: Gender differentials and structural changes of proprietorship in the Indian unorganized enterprises. *Global Business Review*, *24*(2), 295–316. https://doi.org/10.1177/0972150919895090

Appendix

Table 5: PLFS 2023–24: average daily wage of casual workers and average 30-day gross earnings of self-employed women in current weekly status by quarter (Rs), estimated from unit data and as published

Statistic	Quarter	Area	Published	Estimate	Difference
Women, casual wage per day	Q1	India	293	293.8	0.8
Women, casual wage per day	Q1	Rural	289	289.1	0.1
Women, casual wage per day	Q1	Urban	345	350.2	5.2
Women, casual wage per day	Q2	India	293	295.1	2.1
Women, casual wage per day	Q2	Rural	287	288.8	1.8
Women, casual wage per day	Q2	Urban	348	355.5	7.5
Women, casual wage per day	Q3	India	293	293.8	0.8
Women, casual wage per day	Q3	Rural	286	286.8	0.8
Women, casual wage per day	Q3	Urban	358	360.4	2.4
Women, casual wage per day	Q4	India	306	307.3	1.3
Women, casual wage per day	Q4	Rural	299	299.8	0.8
Women, casual wage per day	Q4	Urban	364	368.8	4.8
Men, casual wage per day	Q1	India	439	439.2	0.2
Men, casual wage per day	Q2	India	451	450.8	-0.2
Men, casual wage per day	Q3	India	451	451.7	0.7
Men, casual wage per day	Q4	India	459	459.7	0.7
Women, self-employed earnings, 30 days	Q1	India	5297	5290.6	-6.0
Women, self-employed earnings, 30 days	Q2	India	5384	5357.6	-26.0
Women, self-employed earnings, 30 days	Q3	India	5503	5499.4	-4.0
Women, self-employed earnings, 30 days	Q4	India	5803	5838.8	36.0

Table 6: Enterprise surveys: estimates from the unit data against the published figures. ASUSE coverage excludes residential renting (NIC 68108); GVA refers to market enterprises. Sources: ASUSE 2023-24 factsheet; NSS 73rd round key indicators.

Statistic	Estimate	Published	Difference (%)
ASUSE 2023-24: establishments (crore)	7.34	7.34	0.00
ASUSE 2023-24: workers (crore)	12.06	12.06	0.00
ASUSE 2023-24: female proprietary, % of establishments	26.17	26.20	-0.11
ASUSE 2023-24: annual GVA (Rs crore)	1,797,277.55	1,797,278.00	0.00
ASUSE 2023-24: annual GVA per worker (Rs)	149,742.05	149,742.00	0.00
ASUSE 2023-24: annual GVA per establishment (Rs)	245,686.76	245,687.00	0.00

Statistic	Estimate	Published	Difference (%)
NSS 73: enterprises (crore)	6.34	6.34	-0.01
NSS 73: female proprietary, % of enterprises	19.54	19.50	0.23
NSS 73: female proprietary, % of OAEs	22.31	22.30	0.05
NSS 73: female proprietary, % of establishments	4.82	4.80	0.45
NSS 73: annual GVA (Rs crore)	1,152,338.82	1,152,338.00	0.00
NSS 73: annual GVA per worker (Rs)	103,743.54	103,744.00	0.00
NSS 73: annual GVA per enterprise (Rs)	181,908.06	181,908.00	0.00